

## 2-Mark Questions & Answers — MPCA Unit 1: Overview of Computer Architecture & Organization

### Q1. Differentiate between Computer Architecture and Computer Organization.

| Feature         | Computer Architecture                                       | Computer Organization                                                |
|-----------------|-------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Focus</b>    | High-level logical aspects visible to the programmer (ISA). | Hardware transparent operational units and their interconnections.   |
| <b>Examples</b> | Instruction set, addressing modes, data types, registers.   | Control signals, memory technology, bus interfaces, clock frequency. |

### Q2. What is the Von Neumann bottleneck?

The **Von Neumann Bottleneck** refers to the throughput limitation caused by the shared single system bus between the CPU and memory for both instruction fetches and data transfers. Because instructions and data cannot be accessed simultaneously over the same bus, the CPU is forced to wait (CPU stall), constraining overall processing speed.

### Q3. List any four performance measures of a computer system.

1. **Clock Cycle Time / Frequency (GHz)**
2. **Instruction Execution Rate (MIPS - Million Instructions Per Second)**
3. **Floating-Point Speed (FLOPS - Floating Point Operations Per Second)**
4. **Execution Time / CPU Response Time**

### Q4. Define MIPS and FLOPS.

- **MIPS (Million Instructions Per Second):** A measure of CPU processing speed defined as:  
$$\text{MIPS} = \text{Instruction Count} / (\text{Execution Time} \times 10^6)$$
- **FLOPS (Floating-Point Operations Per Second):** A metric measuring hardware performance in numerical, scientific, and AI floating-point calculations (GFLOPS, TFLOPS, PFLOPS).

### Q5. What is meant by instruction set architecture (ISA)?

**Instruction Set Architecture (ISA)** is the abstract boundary/interface between computer hardware and software. It defines everything a programmer or compiler needs to know to execute code on a CPU, including supported instructions, data types, register sets, memory addressing modes, and interrupt handling models.